Nicolás Bodnariuk

Data Scientist

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https://nbodnariuk.github.io

Profile

Data scientist with a PhD in ocean and atmospheric sciences, specialized in applied analytics, uncertainty quantification, predictive modeling and geospatial data processing. Experienced working with real-world large environmental datasets, numerical modeling workflows and statistical analysis. Skilled at turning complex data into clear, actionable insights through quantitative analysis and effective data storytelling. Comfortable operating across research and business-facing environments, including private-sector consulting.

Technical Skills

Programming	Julia, MATLAB, Python
Data analysis	clustering, large-data set statistical analysis, time series analysis, spatial field pattern extraction
Modeling	Monte Carlo simulations, stochastic modeling, predictive analytics, dynamical systems modeling

Professional Experience

2021–Present **Independent Consultant (Environmental Analytics)**, *Bahitek SRL*, Buenos Aires, Argentina
- Built Monte Carlo uncertainty propagation workflows integrating HYSYS and MATLAB to quantify output variability in industrial pumping networks.
- Processed meteorological data streams (Python, Julia) for pollutant dispersion modelling used by Profertil, Mega, Renova, Viterra and TGS.
- Performed statistical analysis of operational data identifying drivers of temperature-humidity anomalies impacting industrial operations (Aluar plant).

2024–Present **Postdoctoral Researcher (Data-driven modelling)**, *CNRS – LMD*, Paris, France
- Developed a fully custom Quasi-Geostrophic model in Julia to simulate ocean dynamics and analyse nonlinear regime transitions using topological methods.
- Processed multi-terabyte Earth observation datasets to identify climate variability patterns.

2016–2024 **Doctorand and postdoctoral researcher (Big data)**, *CONICET*, Buenos Aires, Argentina
- Applied statistical methods to detect change points, spatiotemporal trends and climate drivers affecting regional ocean circulation on interannual timescales.

Education

2022 **Ph.D. Atmospheric and Oceanic Sciences**, *University of Buenos Aires*

2015 **M.Sc. Physical Oceanography**, *Universidad Nacional del Sur*

Scientific Publications

Author of more than 15 peer-reviewed articles in climate and environmental sciences, applying large-scale data processing and quantitative analysis. Full list available at nbodnariuk.github.io.

Communication and Storytelling

Extensive experience teaching and communicating complex concepts in academic environments, with demonstrated ability to present results using data-driven storytelling and clarity tailored to diverse audiences.

Languages

Spanish (native), English (C2), French (C1), German (B1)